

Unified Final Project Plan (Complete)

SIH26231 — Digital Companion for Field Drug Testing

Mobile-first presumptive testing, tamper-evident records, Panchanama-ready documentation and full technical architecture

Project vision: Build an Android-first application, with an optional supervisor web dashboard, that works with existing colourimetric field-test kits. It converts a subjective colour change into a calibrated, confidence-aware presumptive result and prepares a Panchanama support PDF for officer review. It does not replace laboratory confirmation or departmental/legal procedure.

This document combines: problem requirements, product scope, officer workflow, Panchanama fields, capture/calibration, timestamp/GPS, tamper evidence, security, supervisor search, tech stack, data model, build priorities and demo script.

1. Exact Problem Requirements

Requirement	Implementation
Existing kit compatibility	No new chemical kit or hardware. Use the existing colourimetric kit with a phone camera.
Image capture	Guided camera capture with reaction area and reference colour card visible in the same frame.
Lighting calibration	Detect card patches and correct illumination/white balance before analysis.
Classification	Return Positive, Negative or Inconclusive using defined, versioned criteria.
Digital record	Bind timestamp, GPS location, operator ID and cryptographic image hash to the test.
Tamper evidence	Create signed, hash-linked records and provide integrity verification.
Searchable log	Filter by test/case ID, date, operator, kit/card serial, result and location.
Scope limitation	Every screen and export states: presumptive result; laboratory confirmation remains necessary.

2. Product Architecture

- **Mobile app (primary):** Flutter Android app for capture, local analysis, GPS, secure key storage, offline records and PDF generation.
- **Supervisor web dashboard (optional):** search, review, export approval, audit viewing and QR-based record verification.
- **Core stack:** Flutter, OpenCV/ArUco, encrypted SQLite, SHA-256, Android Keystore/ECDSA, Firebase sync/auth/storage. Use deterministic colour analysis as the primary result engine; ML can be an additional research feature.
- **Design rule:** fail closed. Missing card, poor image, invalid timing, GPS uncertainty or absent signature should produce flagged/inconclusive—not a confident-looking result.

3. Complete Officer Workflow

1. Officer authenticates using login plus PIN/biometric confirmation.
2. Create/select case and seizure/test ID; scan reference-card/kit serial if available.
3. Enter preliminary seizure information: place, package/sample number, kit type and witness details.
4. Open guided camera; align reaction zone and reference card inside overlay.
5. App rejects poor conditions: missing card, blur, glare, over/under exposure, incorrect angle or invalid reaction-time window.
6. Capture valid image. App automatically records timestamp and GPS; it keeps an original evidence image plus a visibly stamped derivative.
7. Calibrate colour, inspect the reaction ROI and display Positive/Negative/Inconclusive with confidence and quality reasons.
8. Record inventory, weights, packaging, samples, seals, Panch witnesses and custody details.
9. Generate signed test record, append it to the local hash chain and save offline.
10. Generate a Panchanama support PDF; supervisor reviews/signs according to department process.
11. Sync metadata/hash first and full media later when connectivity is available.

4. Panchanama-Ready Data Module

The application must generate a **department-configurable Panchanama support draft**, not declare itself to be the final legal Panchanama. Officers and supervisors must review and execute the official form/process required by their agency.

Section	Fields to include in app and exported PDF
Case identity	Department/unit; case/FIR/NCB number; document version; seizing officer name/rank/ID; supervisor; app/model/calibration version.
Place and time	Address/landmark; latitude/longitude; GPS accuracy; capture time; UTC/timezone; device/network time status; search start/end time.
Search context	Authority/warrant/reference where applicable; controlled legal/SOP reference list; person, premises or vehicle searched; owner/occupier/possessor details.
Panch witnesses	Witness name, address, contact/ID, local/independence declaration, presence time, signature/acknowledgement, refusal/absence reason.
Possessor details	Name/identity; recovery location; presence during classification/weighing/packing; acknowledgement or refusal note.
Seized inventory	Package/item number; description/form/markings; gross/net/tare weight; quantity; found-at location; image references; barcode/QR label.
Field test	Kit/batch/serial and expiry if available; sample tested; test start/end; reaction window; calibrated outcome; confidence; inconclusive reason; operator.
Sampling and sealing	Sample mark; quantity; duplicate sample; packaging method; seal number/description; seal image; slip/signature status; seal-intact check.
Custody and forwarding	Handover events; sender/receiver; time; parcel count; seal condition; godown/malkhana reference; test memo/lab dispatch/receipt reference.
Attachments	Original/stamped images; scene, package and seal images; map; inventory; signature pages; generated PDF hash and QR verification.

Official NDPS materials indicate that classification, weighing, packaging and numbering should be conducted/documentated in the presence of search witnesses and the person from whose possession the substance was recovered; inventory, samples and seal information are also central to the seizure record. [web:32][web:36][web:49][web:50]

5. Capture, Calibration and Classification

- Use a printed colour card with known colour/greyscale patches, four ArUco markers, unique serial and QR code.
- Live quality gate checks card detection, blur, glare, exposure clipping, perspective and reaction-time validity.
- Rectify the card, correct illumination, convert corrected reaction pixels to CIE Lab, and compare them with kit-specific reference colours using ΔE .
- Use versioned Positive/Negative/Inconclusive thresholds. Keep the deterministic method explainable; add ML only as an auxiliary model.
- Capture 3–5 frames; select the best quality image and flag inconsistent frames or possible replay/photo-of-photo patterns.
- Show both result confidence and capture-quality status. Low confidence must map to Inconclusive with a clear retake/confirmation instruction.

6. Automatic Timestamp and Geolocation

- At accepted shutter capture, write timestamp, UTC value, timezone and device time-source status to canonical metadata.
- Acquire GPS immediately before/after capture and retain latitude, longitude, accuracy, provider and location acquisition time.
- Automatically add a human-readable timestamp and geolocation label to the **evidence derivative**. Preserve the original image separately and hash both files.
- If GPS is unavailable or too inaccurate, create a location_unconfirmed flag and require an explanation/attestation. Never silently leave it blank.
- Keep test time, upload time and any subsequent edit/review time distinct in both app and PDF.

7. Tamper-Evident Evidence and Security

Layer	Implementation
Original image	SHA-256 of raw image bytes; preserve original capture separately from any stamped/display version.
Metadata binding	Hash canonical JSON containing result, timestamp, GPS, operator, card/kit ID, version data and image hashes.
Hash chain	Each record includes previous_record_hash; editing, deleting or reordering prior records becomes detectable.
Signatures	Hardware-backed device key plus officer-controlled signature/PIN. This demonstrates provenance, not absolute truth of the event.
Offline-first	Encrypted local storage; records/signatures work without internet; metadata/hash sync before large image uploads.
Independent anchor	Optional SMS hash anchor or periodic server/Merkle root as an enhancement.
Verifier	QR/page recomputes hashes and validates signatures; displays Verified, Modified or Incomplete.
Audit and exports	Log views, corrections, export and approvals. Redact by default; watermark approved external exports.

8. Panchanama Support PDF Output

Generate one consolidated, pagination-safe PDF package. Before supervisor approval, mark it clearly as **Draft / For Review**.

- Cover page with case/test ID, status, generated-at time and QR verification code.
- Presumptive field-test page: result, confidence, kit/card data, tested sample/package and operator.
- Image evidence page: original reference, visibly stamped evidence image, timestamp, coordinates, GPS accuracy and location description/map.
- Panchanama support pages: search/seizure context, witnesses, possessor, inventory, weights, samples, seals and chronological narrative.
- Chain-of-custody page: handovers, dates/times, seal condition and forwarding/lab references.
- Integrity certificate: image, derivative, record and PDF hashes; previous hash; device/officer signatures and verifier status.
- Mandatory disclaimer: “Presumptive field-test result only. Laboratory confirmatory testing and applicable departmental/legal procedure remain necessary.”

9. Roles, Log and Dashboard

Role	Access
Officer	Capture, enter field data, sign and view own records; generate draft PDF.
Supervisor	Review, approve configured actions, co-sign selected records and approve exports; view unit records.
Auditor	Read-only access to records, signature/hash chain and audit logs.
Admin	Provision/revoke officers, devices, cards and templates; separate case-content access where possible.
Lab reviewer	View field-test/sample information and add lab reference/result without altering original field record.

Search filters: case/test ID, date range, officer, unit, location/radius, outcome, inconclusive reason, kit/batch, card serial, sync status, seal status and lab-confirmation status.

10. Tech Stack

Layer	Recommended choice	Why
Mobile app	Flutter + Dart (Android-first)	One codebase; fast prototype; good camera/GPS/keystore access
Computer vision	OpenCV + ArUco	Card detection, blur/glare checks, colour calibration
Classification	CIE Lab + Delta E (deterministic)	Explainable, courtroom-friendly logic
Optional ML	TensorFlow Lite	Comparison/auxiliary model only
Local DB	SQLite + SQLCipher	Encrypted offline storage
Backend	Firebase Auth + Firestore + Storage + Functions	Auth, role rules, sync, scalable search
Cryptography	SHA-256 + ECDSA P-256	Hashing and digital signatures
Key storage	Android Keystore / StrongBox	Hardware-backed keys where available
GPS	geolocator plugin	Location + accuracy + provider
PDF	Flutter pdf package	Panchanama-support PDF generation
Maps/geocoding	OpenStreetMap / Nominatim	Human-readable location
QR scan	Google ML Kit	Scan officer ID, card serial, case barcodes

11. How Information is Verified

- **Image/test quality:** card presence, blur, glare, exposure, reaction-time window, multi-frame selection.
- **Colour classification:** store raw colour, corrected Lab, Delta E, threshold version, model version, confidence and quality score.
- **Timestamp:** capture time, GPS time, signing time, upload time; flag clock mismatch; never overwrite original test time.
- **GPS:** lat/long, accuracy, provider, time, mock-location flag; mark location_unconfirmed when needed.
- **Operator identity:** login ID, device ID, PIN/biometric confirmation, optional badge QR scan; device signature + officer signature.
- **Integrity:** image hash, metadata hash, record hash, previous hash, signature status, chain verification.

12. Security from Outside Users

Layer	Measures
Device	Officer login + PIN/biometric; encrypted local DB; Android Keystore; FLAG_SECURE; auto-lock; remote-wipe concept.
Network	HTTPS/TLS everywhere; Firebase Auth tokens; no admin keys in app; private storage paths; short-lived signed URLs.
Database	Role-based Firestore rules; officers see own records; supervisors see unit records; auditors read-only; admins provision but not auto-view case data; prevent overwrite of signed records; full audit log.
Data leak	Redact by default; supervisor approval for bulk export; export watermarks; log every export; QR verification instead of open links.

13. Supervisor Search

Supervisor dashboard should support search by: case/test ID, officer, date range, result, location/radius, kit/batch, card serial, custody status, seal status, panchanama status and lab status. Each opened record shows summary, evidence image, Panchanama data, integrity panel and actions (review, co-sign, approve export, download report).

14. Suggested Firestore Data Model

Collection	Key fields
users	user_id, role, unit_id, assigned_device_ids, status
devices	device_id, officer_id, public_key_fingerprint, status, last_seen
reference_cards	card_serial, issued_to, calibration_version, status
tests	test_id, case_id, officer_id, unit_id, timestamp_utc, geo, result, confidence, image_hash, record_hash, previous_hash, signature_status, panchanama_status, sync_status
test_media	test_id, original_image_path, stamped_image_path, generated_pdf_path
custody_events	event_id, test_id, sender, receiver, timestamp, seal_status
audit_logs	actor_id, action, record_id, timestamp, previous_audit_hash, audit_hash

15. Build Priorities

Priority	Build item	Why it matters
P0	Guided capture, card-in-frame check, automatic timestamp/GPS, local record and searchable log.	Meets the literal problem requirements.
P0	Calibration, reaction ROI, deterministic classification and inconclusive handling.	Demonstrates objective colour interpretation.
P0	Image hash, signed record, hash chain and simple verifier.	Demonstrates tamper evidence.
P0	Panchanama support fields plus PDF generator.	Connects the solution to actual narcotics documentation.
P1	Encrypted offline store and staged Firebase sync.	Makes it usable in weak-connectivity field conditions.
P1	Supervisor review, roles, audit events and approval gates.	Demonstrates deployment maturity.
P2	Burst capture, QR registry, SMS/Merkle anchoring, watermarking and multilingual UI.	Adds differentiation if time permits.

16. Live Demo Sequence

1. Create a case, add two Panch witnesses and enter package details.
2. Show a poor capture attempt; app rejects it due to blur, glare or missing card.
3. Capture a valid test image; visibly show automatic timestamp and GPS overlay.
4. Show calibration and classification with confidence and an Inconclusive pathway.
5. Open the signed evidence record with hashes, signatures and chain reference.
6. Search saved tests by case ID.
7. Generate the Panchanama support PDF and show the witness, inventory, seal, test, location and integrity sections.
8. Modify a copy of an image or metadata and show verification failure.
9. Close with the presumptive-result disclaimer and lab-confirmation handoff.

17. Boundaries and Final Pitch

- Do not claim laboratory-grade certainty, automatic legal admissibility or a universally valid Panchanama.
- Use department-configurable forms and obtain the concerned department's review for exact fields, templates, retention and signatures.
- Do not overwrite edits; store amendments with actor, reason, timestamp and prior value.
- The tool supports evidence documentation; it does not independently prove physical truth, legal compliance or substance identity.

Final pitch: "A calibration-first, offline-capable digital evidence companion that captures, classifies and cryptographically verifies presumptive field-test results, while automatically assembling timestamped, geolocated testing, seizure, witness, inventory, seal and custody details into a Panchanama-ready review package."

BNSS materials describe witness-signed seizure lists and audio-video recording of search processes; NDPS rules/materials describe inventory, sampling, witness presence and seal-related processes. These are design inputs only—the final implementation must follow the competent department's SOP and legal advice. [web:33][web:34][web:47][web:48]